Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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Contents

Chapter 1

Introduction

Chapter 2

Literature Review

2.1 Biomedical Information Extraction

2.1.1 Definition

Biomedical IE is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system: extracted entities and relations are later assembled into a Knowledge Graph (KG) and validated against biomedical ontologies **[NEED REFERENCE]** (section)

2.1.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with an analysis by Lu et al. (2011) demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data - a 2024 study mapping over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to a level where the volume of biomedical publications has made it difficult for researchers to keep track of new literature [2]. Literary analysis is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and a substantial majority of literature goes unseen that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this thesis, as a corpus that is both large and continuously updated is exactly the setting in which static benchmark-trained pipelines are most likely to be used.

2.1.3 Tasks

Biomedical IE is usually comprised of several sub-tasks: Named Entity Recognition (NER), which identifies and stores types of spans of text referring to biomedical concepts, and Relation Extraction (RE), which identifies semantic relationships between recognised entities, and Named Entity Normalisation (NEN), which maps extracted mentions to pre-defined identifiers from an existing ontology [4, 5].

Unlike NER, which only labels a span of text as belonging to a given entity type, NEN maps that span to an established identifier from a pre-defined ontology or controlled vocabulary, storing which specific concept the span is referring to, rather than what kind of concept it is. This resolves terminology ambiguity challenges discussed in the following section.

2.1.4 Challenges

Long or complex sentences, which are common in biomedical abstracts, summarising multiple findings into a single sentence, can cause a significant challenge for these tasks [6]. This is particularly prominent for relation extraction, which typically requires attention over the text's distance between the two entities within a sentence: if two entities are far apart in this text, multiple clauses and mentions of competing entities can make it more difficult to correctly associate the entities with their corresponding relation [7], increasing the risk of both False Negatives (FNs) and False Positives (FPs). Sequential relation extraction models have been shown to have difficulty capturing these long-range dependencies, resulting in poorer performance on longer sentences [8]. The information density of biomedical literature may partly reflect conventions of scientific communication, which emphasise conciseness, clarity and precision of information [9].

Beyond task complexity, biomedical text includes domain-specific challenges that separate it from general IE: dense and evolving terminology [10, 11], and high rates of entity name ambiguity [12], where the same underlying concept is frequently expressed through several distinct strings, such as a drug's brand name versus its generic name, or a gene's full name versus a locally-defined abbreviation within a specific paper, while on the other hand, a single string can refer to different concepts depending on context. As a result, extracted entities referring to the same concept would be treated as distinct, fragmenting information than consolidating it. NEN aims to address this, mapping ambiguous or synonymous mentions onto a shared established identifier. This task is largely important for subsequent KG construction, as graph nodes rely on identifiers to be queryable and to allow separate mentions of the same concept to be merged into a single node **[NEED REFERENCE]**.

In addition to these domain-specific challenges, significant methodological limitations compound the difficulty of RE evaluation. Meaningful comparison between biomedical IE systems requires consistent evaluation frameworks: differences in annotation guidelines, entity and relation identifiers, and train/test splits across benchmark datasets can make performance difficult to compare directly between studies, even when the underlying literature appears homogeneous **[NEED REFERENCE]**. Separately, strong generalisation across literature distributions requires models to perform reliably across heterogeneous literature - literature differing in general publication date, subdomain, terminology, and reporting style from that of what a system was developed and evaluated on **[NEED REFERENCE]**. Benchmark datasets are fixed at the point of annotation, while biomedical literature continues to grow and evolve, therefore, strong performance on a static, in-distribution benchmark therefore does not offer any guarantee of comparable performance on literature published after the benchmark was constructed, or on subdomains that may have been underrepresented within it. This distinction between benchmark-reported and generalised performance is further discussed in Section **??**, and motivates evaluating the frozen extraction pipeline developed against a separately constructed contemporary corpus.

2.1.5 Traditional Methods and Evaluation

Before generative LLMs became viable for biomedical IE, the field progressed through several distinct systems. Initial systems relied on manually-constructed rules and biomedical ontologies, matching text against predefined patterns and dictionaries to identify entities and relations **[NEED REFERENCE]**. These were gradually improved by and progressively replaced by traditional machine learning methods, which used syntactic and contextual features to train classifiers such as support vector machines for NER and RE **[NEED REFERENCE]**. Neural systems were then gradually implemented, initially through architectures such as recurrent and convolutional neural networks trained on word representations and later through transformer-based Pretrained Language Models (PLMs) fine-tuned specifically on biomedical text, such as BioBERT and PubMedBERT **[NEED REFERENCE]**. The progression of pretraining on biomedical corpora and task-specific fine-tuning allowed PLMs to substantially outperform earlier feature-based methods, resulting in them becoming the main extraction-system approach for specialised biomedical IE prior to later prompt-based generative approaches.

Development and evaluation throughout this progression have mainly been driven by benchmark datasets, with models commonly trained and evaluated on fixed, established corpora. However, relying on individual benchmarks introduces comparability and generalisability limitations outlined previously. Recent work has therefore aimed to address these limitations through more consistent evaluation frameworks and broader heterogeneous training data: Sanger et al. (2025) compared multiple PLMs before evaluating knowledge extraction of the best-performing model across five datasets within a consistent evaluation framework

[13]. Similarly, Lai et al. (2023) introduced BioREx, which consolidates heterogeneous biomedical RE datasets to support broader training and improve generalisation [14].

Both of these approaches, however, rely on PLMs tuned on labelled benchmark data, with performance assessed against fixed, in-distribution, annotated datasets. This highlights an important methodological limitation for practical application, where extraction systems must perform robustly on dynamic, contemporary literature, rather than a static benchmark corpus. Therefore, the extent to which performance established on benchmark data transfers to unseen contemporary literature remains an important question and a motivator for Research Question 2 (RQ2).

2.2 Large Language Models for Biomedical Information Extraction

2.2.1 Transition from Specialised Models to LLMs

More recently, LLMs have driven a broader shift in knowledge extraction and KG construction from rule-based and module pipelines towards generative and adaptive frameworks [15]. Early evidence supports competitive extraction performance with Zhang et al. (2024) reporting an F1 score of 0.84 for GPT-4 on biomedical RE for the Genetic Association Database (GAD) benchmark dataset, with GPT, BioBERT, and PubMedBERT performing comparably on the GAD dataset under certain extraction system frameworks [16]. However, it is important to note that GPT-3.5 and BioBERT v1.1 performed considerably worse than PubMedBERT on the masked ChemProt dataset, suggesting that relative model performance may vary substantially across biomedical RE tasks and datasets.

Additionally, performance gains from LLMs are not limited to biomedicine. Evaluations of LLMs for KG construction have observed an F1 score of 99.35% for Meta's LLaMA 3 model on OpenStack system log data [17], indicating extraction capability is not domain-specific.

This body of evidence motivates the pipeline design adopted in this project: an LLM-based extraction approach using structured JSON outputs and few-shot prompting, developed on an existing benchmark, BioRED [18], before evaluation on a newly-assembled contemporary corpus.

2.2.2 Prompt-Based and Few-Shot Extraction

Unlike specialised pretrained language models such as BioBERT and PubMedBERT, which require task-specific fine-tuning on labelled data to perform NER or RE, generative LLMs can perform biomedical IE directly through prompting, without any parameter tuning [19]. This approach can allow flexibility through natural-language instructions, rather than requiring additional model training. As a result, a single general-purpose model can potentially be adapted to different entity types, relation schemas, and extraction tasks through only prompt refinement **[NEED REFERENCE]**.

Several approaches can be implemented: zero-shot prompting - where the model is given only a natural-language description of the task, few-shot prompting - where the prompt includes structured examples of exemplary outputs corresponding to given inputs, and schema-guided instruction, where the prompt specifies the exact entity and relation types that the model should extract [20, 21, 22].

The extent to which an LLM is able to compete with fine-tuned extraction systems depends heavily on how the prompt itself is constructed. Recent comparisons between zero-shot and few-shot prompting on biomedical IE tasks suggest that including examples in the prompt can improve extraction quality, as can annotation guidelines or provided schema within the prompt itself [21]. However, these gains are not consistent across biomedical IE tasks: Zhang et al. (2024) observed only limited improvements from one-shot prompting on some datasets, with performance decreasing on the ChemProt dataset. Given ChemProt's more complex relation schema [23], these findings suggest that the effectiveness of few-shot prompting may depend on task complexity and how well the selected examples represent the target annotations **[NEED REFERENCE]**. This prompt-dependency is important for this project specifically as it provides a basis for developing and iterating on multiple Extraction Pipeline Iterations (EPI) during BioRED-based development, rather than treating prompt design as a single, fixed choice that remains unexamined.

2.2.3 Performance and Limitations

Despite this flexibility, LLM-based extraction introduces failures that are usually absent from, or less significant in fine-tuned supervised models **[NEED REFERENCE]**. The most widely discussed is drawing from domain knowledge: LLMs may produce extracted entities or relations that are plausible but not present in the supplied text [22]. This is particularly important for biomedical IE, where an extracted relation is only useful if it is grounded in the given literature, rather than drawn from existing background knowledge.

Problems also exist with LLM consistency and control: LLM outputs can vary across repeated runs on the same input [24], and extraction quality has been shown to be sensitive to changes in prompt wording or the specific examples chosen for few-shot prompting [16, 21]. Models may also fail to strictly follow a predefined relation schema, extracting relation types outside the intended ontology or omitting required output fields, which can complicate downstream pipelines that assume a fixed schema [24]. Together, this behaviour may create a precision-recall trade-off where more exhaustive extraction increases recall at the cost of precision, while more conservative prompting tends to do the opposite.

2.2.4 Structured Output

An important design consideration is the format in which extracted information is returned. Limiting LLM output to a structured, machine-readable format, such as JSON conforming to a predefined schema, rather than unstructured text generation is intended to make extraction results easily usable in downstream processing without requiring separate parsing steps [22]. For a pipeline that extracts entities and relations that are subsequently assembled into a KG (Section 2.3), this is an important methodological choice: KG construction requires each extracted entity and relation to be reliably identifiable as a discrete unit whereas unstructured free-text output requires an additional extraction from the output extractions. Structured output generation is therefore treated in this project as a component of the extraction pipeline itself, rather than a post-processing step.

2.2.5 Implications for Biomedical Information Extraction

In summation, the current literature provides a more nuanced view than the conclusion that LLMs perform interchangeably with specialised biomedical extraction models. LLMs reduce the dependence on task- and subdomain-specific fine-tuning and offer flexible, adaptable extraction across entity and relation types, which is an important practical advantage over supervised PLM pipelines discussed in Section 2.2.1. However, this flexibility comes with significant caveats: extraction performance that is sensitive to prompt construction and appears to vary across datasets and specific tasks while the generative aspect of these models comes with reliability concerns such generation of information not grounded in the source text, variable outputs, and schema non-compliance. For a pipeline intended to generalise from a benchmark corpus to unseen contemporary literature, this distinction is important: benchmark-evaluated precision and recall capture extraction accuracy on in-distribution text, but do not directly capture whether a frozen prompt-based pipeline remains robust when applied to literature it was not developed against. This motivates evaluating not only whether extraction quality transfers from BioRED to contemporary PubMed abstracts, but whether the specific points of failure discussed become more or less significant during generalisation.

2.3 Knowledge Graph Construction Using LLMs

2.3.1 Knowledge Graph Representation

A KG represents information as a set of entities, or nodes, connected by relations, or edges, with both nodes and edges, sometimes including additional information such as a confidence score, or source identifier **[NEED REFERENCE]**. The fundamental unit of a KG is the triple, which represents a single relationship between two entities. For example, the finding that warfarin is associated with intracranial haemorrhage can be represented as the triple:

$$(\text{Warfarin, Association, Intracranial Haemorrhage}).$$

A KG is fundamentally a collection of triples, with shared entities linking separate triples into a connected structure rather than a set of isolated relations.

This structure is well suited to biomedical information where individual findings are often reported in isolation within a single abstract but gain insight when connected across the literature. Representing extracted findings as a knowledge graph connects mentioned of the same entity across different papers into a single queryable structure. This can reveal clusters of relations and indirect connections that would not be apparent from a single abstract in isolation. For example, identifying an indirect association between two entities that are never mentioned together in the same source text, but are connected to each other through a mutual entity **[NEED REFERENCE]**.

This representation maps naturally onto the information extraction tasks introduced in Section 2.1. NER identifies the spans of text that become candidate nodes in the graph and NEN maps those candidate nodes to shared predefined identifiers, ensuring that semantically separate mentions of the same concept are consolidated into a single node rather than represented as duplicates. Finally, RE identifies the edges connecting these nodes. The insights KGs provide are precisely why NEN is a key component of the extraction pipeline: if duplicate mentions of a single concept were not resolved, biomedical concepts would become fragmented across several nodes, undermining the purpose that KGs are intended to provide.

2.3.2 LLM-Based Knowledge Graph Construction

Section 2.2.1 outlined how the recent LLM-driven work represents a broader shift in knowledge extraction from rule-based and modular pipelines towards adaptive generative frameworks [15]. In the context of KG construction, this shift changes the overall architecture of the entire extraction-to-graph system. A traditional, modular architecture treats NEN, NER, and RE as separate stages, each performed by a distinct model or tool, before the resulting entities and relations are assembled into a graph **[NEED REFERENCE]**. Although the exact ordering of these steps may differ, an example module workflow is:

Text → RE → NER → NEN → KG **[NEED REFERENCE]**.

An LLM-driven approach instead allows a single model to conduct RE and

NER directly from text within one pipeline step, with a separate subsequent step conducting NEN before KG construction:

$$\text{Text} \rightarrow \boxed{\text{LLM joint NER + RE}} \rightarrow \text{NEN} \rightarrow \text{KG}.$$

This consolidation is possible as an LLM with a single adequate prompt can both recognise entity spans and the relations between them in text. Recent work evaluating end-to-end biomedical relation extraction using LLMs demonstrates the potential of joint-extraction [22] (?) and multi-extraction prompting has also been shown to support this within a single framework without requiring additional fine-tuning [24].

Furthermore, because the LLM's output is constrained to an easily-parsable schema, rather than free text, the extracted entities and relations are already in a form that does not require extra parsing or interpretation to be mapped onto graph triples. This is what allows the LLM-driven architecture to abstract several traditionally separate pipeline stages into fewer steps.

It is worth noting that LLMs can support KG construction beyond direct entity and relation extraction, including schema and ontology generation or refinement [15] **[NEED REFERENCE]**. This project primarily uses an LLM-based approach for structured information extraction, while also implementing LLM-assisted error analysis during iterative development of the extraction system. The graph schema and ontology itself, however, remain predefined, rather than being generated or refined by the LLM.

2.3.3 Ontology Integration and Entity Normalisation

Within KG construction, NEN provides the connection between entity mentions extracted from free text and established biomedical concepts. As discussed in 2.1.4, relying on separate nodes formed by unmapped string would fragment mentions of the same biomedical concepts across the graph. Mapping these mentions to ontology identifiers instead allows references to the same underlying concepts to be consolidated.

Biomedical ontologies and vocabularies address this by providing a fixed set of biomedical concepts, each associated with an established identifier, that extracted mentions can be mapped to **[NEED REFERENCE]**. NEN therefore acts as the step that allows a free-text span to be represented by an identifiable graph node.

Beyond identifier assignment, the ontology can also constrain the KG more broadly, restricting the extracted entities and relations to respective predefined, established sets **[NEED REFERENCE]**. This introduces a distinction between normalisation and ontology-based validation. While NEN maps an extracted span of text to a consolidated, established concept, validation can assess whether extracted entities and relations align with existing ontologies. Ontology integration can therefore support the consolidation and validation of the resulting KG

[NEED REFERENCE].

2.3.4 Challenges and Quality of LLM-Constructed Knowledge Graphs

Because a KG is assembled directly from the outputs of the extraction pipeline, errors made in the NER, RE, and NEN stages are not isolated from other stages, but instead pass into the resulting graph. For example, a false-positive entity or relation will become an incorrect node or edge, while false negatives would become missing edges. False-positive relations may be the more detrimental of the two errors, as the graph explicitly states them as knowledge rather than just excluding them **[NEED REFERENCE]**. Errors at the NEN stage, discussed in Section 2.3.3, have a similar effect: false-negative normalisation results in mentions of the same concept separated across duplicate nodes, while false-positive normalisation consolidates separate concepts into a single node **[NEED REFERENCE]**.

Because of the connected nature of steps in an extraction pipeline system, KG quality is dependent on the quality of the original extraction. A KG can also be evaluated at the graph level for properties such as consistency, completeness, duplication, and compliance with its ontology or schema **[NEED REFERENCE]**.

2.4 Benchmark Datasets and Evaluation

2.4.1 Biomedical IE Benchmark Datasets

Progress in biomedical IE, has largeley been measured against manually annotated benchmark corpora, which provide the ground truth against which NER and RE systems are trained and evaluated **[NEED REFERENCE]**. As annotation is costly and requires domain expertise, researchers typically constrain their annotations to particular biomedical entities, relations, or domains **[NEED REFERENCE]**, targeting a specific set of entity and relation types, rather than biomedical text in general **[NEED REFERENCE]**. Because of this, different benchmarks are not directly interchangeable as an extraction system's reported performance reflects the entities, relations, and schema of the specific benchmark it was evaluated on, rather than a measure of their general biomedical extraction ability. This benchmark heterogeneity was discussed in Section 2.2.1, where performance was shown to vary significantly between the GAD and ChemProt datasets.

This heterogeneity is stated briefly in table **[NEED REFERENCE]**, comparing the domain and entity/relation scopes of three benchmark datasets previously mentioned in this report.

Table 2.1: Performance of biomedical relation extraction models

Dataset	Subdomain	Entity/Relation Type	Project Relevance
GAD [25]	Genetic association	Gene-disease association pairs: single binary relation type	Narrow, single-relation benchmark
ChemProt [23]	Chemical-protein interaction	Chemical and protein entities: multiple interaction subtypes	Benchmark with a more complex relation schema
BioRED [18]	General biomedical abstracts	Multiple entity types (e.g., gene, disease, chemical, species) and multiple relation types	Broader biomedical coverage

This comparison demonstrates that the interpretation of a strongly-performing extraction system is dependent on the dataset it is built on: a model exhibiting strong results on the GAD dataset demonstrates a narrower scope of extraction strength than that of the BioRED dataset as the two differ substantially in terms of how many entity and relation types must be extracted and, therefore, understanding of context from biomedical literature. This difference motivates the choice of BioRED as the benchmark dataset used to develop the extraction pipeline in this project as a benchmark covering multiple entity and relation types is more representative of the broader task of general-purpose KG construction.

2.4.2 BioRED

BioRED, introduced in Section 2.2.1, is a manually annotated corpus of 600 PubMed abstracts, constructed to support document-level, multi-entry, multi-relation biomedical relation extraction **[NEED REFERENCE]**.

The corpus comprises six entity types: chemical entity (`ChemicalEntity`), disease or phenotypic feature (`DiseaseOrPhenotypicFeature`), gene or gene product (`GeneOrGeneProduct`), sequence variant (`SequenceVariant`), and organism taxon (`OrganismTaxon`). Eight relationship types connect these entities: association (`Association`), bind (`Bind`), comparison (`Comparison`), conversion (`Conversion`), cotreatment (`Cotreatment`), drug interaction (`Drug_Interaction`), negative correlation (`Negative_Correlation`) and positive correlation (`Positive_Correlation`) [18].

In total, the corpus contains 20,419 entity annotations, mapped to 3,869 identifiers **[CHECK INFO]**, where, each entity mention is normalised to an established identifier in the annotation itself [18]. Relations are also labelled as describing either a new finding or existing background knowledge, allowing systems to distinguish novel findings from existing background information. As with the other benchmark corpora discussing in this review, BioRED is split into training, development, and test splits, enabling consistent comparison between systems evaluated on the dataset [18].

Because of these characteristics of the BioRED dataset, BioRED is well suited to the aims of this project. While other commonly-used biomedical RE benchmark datasets are constructed around a single entity or relation type, BioRED's wide scope of multiple entity types and relations and existing entity normalisation align closely with this project's general-purpose extraction task required for KG construction. This scope is the main motivator for its selection as the benchmark dataset used to iteratively develop an extraction pipeline in this project, ahead of evaluation against contemporary literature.

However, BioRED remains as fixed, manually annotated, benchmark-distributed corpus where annotations reflect the 600 abstracts sampled during construction and its entity and relation schema remains fixed rather than reflective of the full scope of biomedical concepts and relationships in contemporary literature. Because of this, while BioRED offers reliable ground truth for developing an extraction pipeline, strong performance on the dataset does not ensure the same performance on literature published after the benchmark was constructed, or on subdomains it was not fine-tuned to represent. This is the same gap introduced in Section 2.1.4 and 2.1.5 and is the main motivator for RQ2 and its relevance to this project: BioRED is treated as a source of reliable data for extraction pipeline development, rather than as evidence of generalised extraction performance, discussed further in Section 2.5.

2.4.3 Extraction-Level Evaluation

2.4.4 Limitations of Benchmark Evaluation

2.5 Generalisation to Contemporary Biomedical Literature

Chapter 3

Methodology

3.0.1 Research Design

3.0.2 Datasets

3.0.3 System Architecture

3.0.4 Information Extraction Pipeline

3.0.5 Pipeline Development

3.0.6 Contemporary Knowledge Graph Construction

3.0.7 Evaluation

3.0.8 Cost Analysis

3.0.9 Software Implementation

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

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